Report 2: UML Class Diagram Report

Project Title: Hotel Booking Management System

Group Members:

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1. Complete Class Diagram Details

Class 1: Guest

Attributs**:**

- guestId: String
- fullName: String
- email: String
- phoneNumber: String

Methods:

+ makeReservation(): void
+ cancelReservation(): void
+ viewHistory(): void

Class 2: Room

Attributes:

- roomNumber: String
- roomType: String
- pricePerNight: double
- isAvailable: boolean

Methods:

+ updateStatus(): void
+ checkAvailability(): boolean

Class 3: Reservation

Attributes:

- reservationId: String
- checkInDate: Date
- checkOutDate: Date
- totalAmount: double
- status: String

Methods:

+ confirmBooking(): void
+ calculateTotal(): double

Class 4: Payment

Attributes:

- paymentId: String
- paymentDate: Date
- amountPaid: double
- paymentMethod: String

Methods:

+ processPayment(): void
+ generateReceipt(): void

2. Relationships and Multiplicities

1.Guest — Reservation

Relationship Type: Association (makes)

Multiplicity: 1 to 0..*

Explanation: One Guest can create zero or many Reservations. Each Reservation is tied to exactly one Guest.

2.Reservation — Room

Relationship Type: Association (includes)

Multiplicity: 1..* to 1

Explanation: A Reservation must include one or more Rooms. A specific Room is associated with one active Reservation at a time.

3.Reservation — Payment

Relationship Type: Composition (has)

Multiplicity: 1 to 1

Explanation: A Reservation requires exactly one Payment. If the reservation is deleted, the payment record is bound to that specific transaction context.